

ES-221 Mechanics of Solids

Stress Analysis of a Circular Torsional Bar

Group No. 10

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Date of Submission: 23/04/2026

1 Objectives

The primary goal of this project is to build a Python-based computational tool that automates the stress analysis of circular shafts subjected to two twisting loads each at a Fixed-Fixed and a Fixed-Free (Cantilever) support. To thoroughly investigate the torsional behaviour of both solid and hollow shafts, our program is designed to:

- Calculate and plot the internal torque (how it varies along the shaft depending on the supports (fixed-free vs fixed-fixed)), maximum shear stress ($\tau_{\max}$) (how that shear stress distributes across the cross section (solid vs hollow circular beam)).
- For any specific cross-section and form of support, plot Mohr's circle to find the principal stresses and use it to calculate stress on a rotated element. This provides a clear, visual breakdown of how stresses transform, highlighting the maximum shear and principal normal stresses.

2 Problem Statement

Consider a circular shaft of length $L = 1000$ mm and an outer radius $R = 30$ mm. We will analyse two variations: a completely solid shaft, and a hollow shaft with $R_{in} = 20$ mm.

Two concentrated torques are applied along its length: $T_1 = 800$ N·m at $z_1 = 350$ mm and $T_2 = 400$ N·m at $z_2 = 750$ mm. The shaft can either be supported at one end only or clamped at both ends. We want to find:

1. How the internal torque $T(z)$ varies along the shaft for each support type.
2. What the shear stress distribution looks like across the cross-section (comparing the solid vs. hollow shafts).
3. The principal stresses and the full stress state at any chosen section and element orientation.

3 Theory and Methods

Statics and Internal Torque

Cantilever (Fixed-Free): Because only one wall is present, the wall reaction must balance all the applied torques. The internal torque $T(z)$ at any section is found by summing the external torques applied to the right of that section (working from the free end inward):

$$T(z) = \sum_{i: z_i > z} T_i \quad (1)$$

This gives a step-function diagram that starts high at the wall and drops each time a load is passed.

Fixed-Fixed: When both walls are rigid, neither end can rotate so the total angle of twist from one wall to the other must be zero:

$$\sum \phi = \int_0^L \frac{T(z)}{I_z G} dz = 0 \quad (2)$$

Applying this compatibility condition gives the left wall reaction:

$$T_A = \frac{T_1(L - z_1) + T_2(L - z_2)}{L} \quad (3)$$

Once T_A is known, the internal torque diagram is built the same way as the cantilever case, just with T_A as the starting value.

Shear Stress and Angle of Twist

The shear stress at any radial position ρ is given by the torsion formula:

$$\tau(\rho) = \frac{T(z)\rho}{I_z} \quad \text{with maximum at } \rho = R: \quad \tau_{\max} = \frac{T(z)R}{I_z} \quad (4)$$

where J is the polar moment of inertia: $I_z = \frac{\pi}{2}(R_{\text{out}}^4 - R_{\text{in}}^4)$ (set $R_{\text{in}} = 0$ for a solid shaft).

The stress varies linearly from zero at the centre to $\tau_{\max}$ at the outer surface, as illustrated in Figure 1.

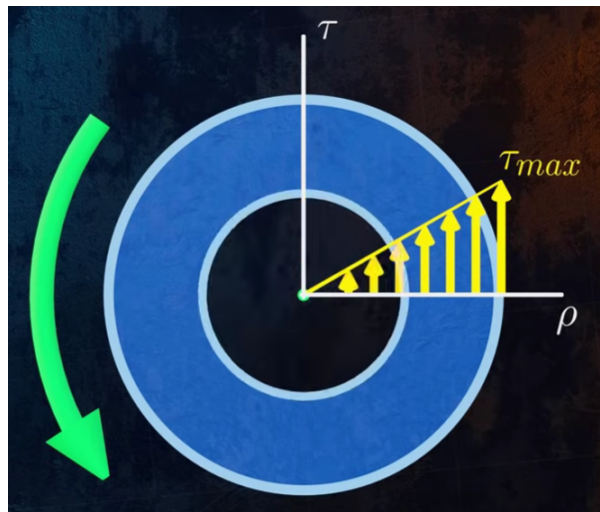


Figure 1: Linear variation of shear stress τ across a hollow circular cross-section. Stress is zero at the inner radius and reaches $\tau_{\max}$ at the outer surface.

The angle of twist at any location is found by integrating the twist gradient:

$$\phi(z) = \int_0^z \frac{T(z)}{I_z G} dz \quad (5)$$

Figure 2 shows how the angle of twist ϕ and the surface shear strain γ are geometrically related on the deformed shaft.

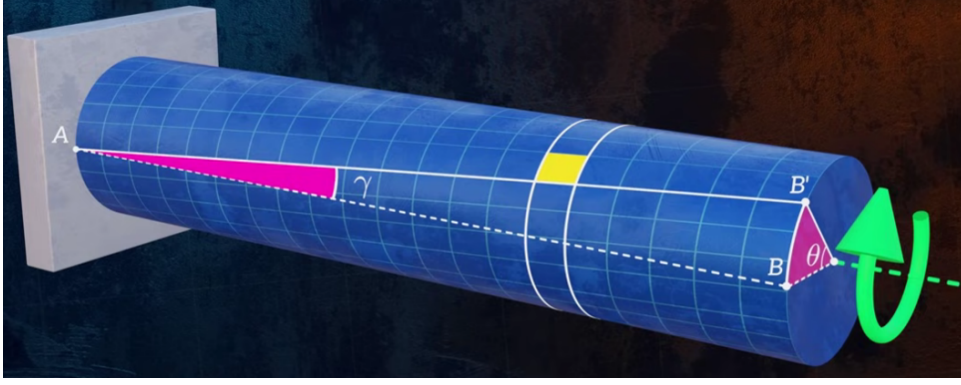


Figure 2: Geometric interpretation of torsional deformation. Point B moves to B' as the shaft twists by angle ϕ , while the shear strain γ describes the angular distortion along the surface.

Mohr's Circle and Stress Transformation

At any surface point under pure torsion, there are no normal stresses only shear: $\sigma_x = \sigma_z = 0$, $\tau_{xz} = \tau_{\max}$. The principal stresses are therefore:

$$\sigma_1 = +\tau_{\max}, \quad \sigma_2 = -\tau_{\max} \quad (6)$$

Mohr's Circle for this state is centred at the origin with radius $= \tau_{\max}$. When you rotate the stress element by angle θ , the stresses on the new faces are:

$$\sigma'_z = \tau_{\max} \sin(2\theta) \quad (7)$$

$$\sigma'_x = -\tau_{\max} \sin(2\theta) \quad (8)$$

$$\tau' = \tau_{\max} \cos(2\theta) \quad (9)$$

4 Numerical Implementation

The analysis was implemented in Python, utilising `numpy` for numerical computations and `matplotlib` for visualisation. The core components are structured as follows:

- **Geometry and Loading:** System parameters (shaft dimensions, torque magnitudes, and load positions) are defined globally. Polar moments of inertia for solid and hollow cross-sections are pre-calculated.
- **Internal Torque:** The shaft length is discretized into 500 spatial points. A loop evaluates the internal torque at each point by sequentially subtracting the applied loads from the initial boundary reaction (e.g., T_A for the fixed-fixed case).
- **Cross-Sectional Stress:** Shear stress, $\tau(\rho) = \frac{T\rho}{I_z}$, is calculated across an array of 300 radial points. Hollow cross-sections are visually shaded in the plots to indicate the absence of core material.

- **Mohr's Circle and Stress Element:** Using $\tau_{\max}$ at the chosen cross-section, the script generates Mohr's Circle and a corresponding 2D stress element. This visualises both the original pure shear state and the transformed normal and shear stresses at any user-defined rotation angle (θ).
- **Analysis Point Selection:** To ensure meaningful stress visualisation, the evaluation point is set to $z = 300$ mm. This guarantees the analysis occurs within an actively loaded region of the shaft rather than at a zero-torque boundary.

5 Results and Discussion

In the **cantilever** setup, the left wall supports the entire 1200 N·m load, making the segment near the wall the most critical.

For the **fixed-fixed** shaft, geometric compatibility distributes the reaction between both walls. Using Equation (3):

$$T_A = \frac{800(1000 - 350) + 400(1000 - 750)}{1000} = \frac{520000 + 100000}{1000} = 620 \text{ N·m}$$

The right wall takes $T_B = 1200 - 620 = 580$ N·m. The maximum internal torque drops nearly in half (from 1200 to 620 N·m), highlighting the structural efficiency of fixed-fixed configurations for carrying heavier loads.

Shear Stress Across the Cross-Section

Figure 3 displays the linear shear stress distribution at $z = 300$ mm, rising from zero at the centre to a maximum at the outer edge.

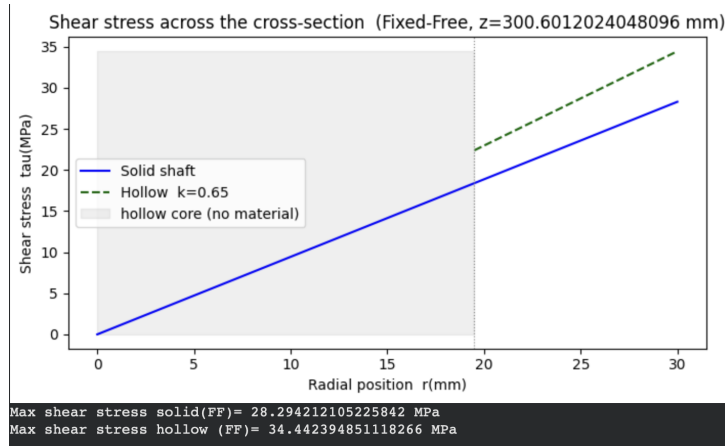


Figure 3: Shear stress across the cross-section at $z = 300$ mm. The hollow shaft has a slightly higher peak stress because it has less material resisting the twist.

Mohr's Circle

For pure torsion, Mohr's Circle (Figure 4) is perfectly centred at the origin with a radius equal to $\tau_{\max}$.

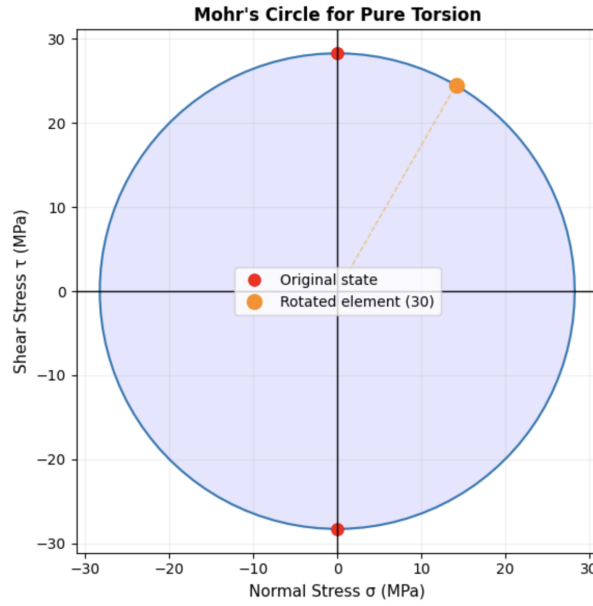


Figure 4: Mohr's Circle for pure torsion at $z \approx 300$ mm. The red dot shows the original state (pure shear), and the orange dot shows the stresses rotated by 30° .

The red dot represents the original pure shear state (zero normal stress). Rotating the physical element by 30° (which moves 60° on the circle) introduces normal stresses, indicated by the orange dot. The principal stresses (maximum pure tension and compression) occur at 45° where the circle intersects the horizontal axis.

Rotated Stress Element

Figure 5 confirms the stress transformation equations mathematically and visually.

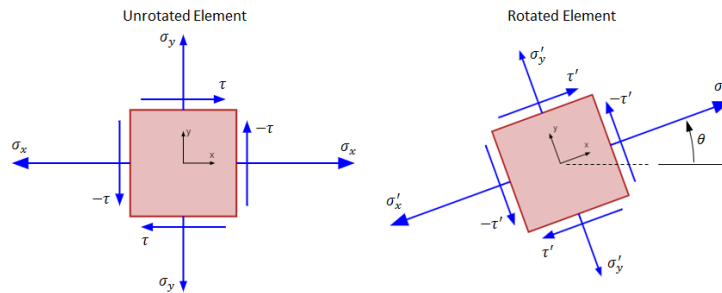


Figure 5: Rotated stress element at $\theta = 30^\circ$ for a section with $\tau_{\max} = 28.3$ MPa. The Normal Stress $\sigma_z = 24.5$ MPa and $\sigma_x = -24.5$ MPa on the complementary face.

Rotating the element by 30° introduces normal stresses of ± 24.5 MPa on opposing faces while reducing the acting shear stress from 28.3 MPa down to 14.15 MPa.

6 Learning Outcomes

- **Determinate vs. Indeterminate Systems:** Successfully solved and compared internal torques for both cantilever (determinate) and fixed-fixed (indeterminate) support conditions.
- **Solid vs. Hollow Stress Distribution:** Demonstrated that shear stress (τ) varies linearly across the cross-section, starting from zero at the exact center of a solid shaft, but starting from the inner

radius (R_{in}) for a hollow shaft.

- **Mohr's Circle Transformations:** Visually proved that changing the orientation angle of a stress element dynamically alters the balance of normal and shear stresses on Mohr's Circle.

References

- [1] Crandall, S. H., Dahl, N. C., & Lardner, T. J. (1999). *An Introduction to the Mechanics of Solids* (2nd ed., Chapter 6). McGraw-Hill.
- [2] Matplotlib Development Team. *Matplotlib: Visualization with Python - Documentation*. <https://matplotlib.org/>
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